Fabrication process for NESS Electrodes

(Note: These steps were initially written by the previous student who designed the fabrication process, I have only modified it based on the changes I made while conducting the process myself.)

Precursor Mixture Preparation

1. Put Spandex gloves on.
2. Weigh CNT on a precision scale, dropping it onto a folded piece of paper. The CNT should constitute 2.4–2.5 wt% of the total composite (accounting for potential incomplete mixing). Silicone oil should be 20 wt% of the total composite. For example, for a 5 g total composite with CNT wt% = 2.4 %: CNT = 0.12 g, silicone oil = 1.0 g, PDMS base = 3.525 g. For 10 g total composite with CNT wt% = 2.5 %: CNT = 0.25 g, silicone oil = 2.0 g, PDMS base = 7.04 g.
3. Add CNT into a glass beaker, including additional 0.001 g to account for losses. Add IPA at a ratio of ~1 mg CNT per 0.5 mL IPA. Keep the beaker sealed with parafilm during the mixing process.
4. Disperse the CNT–IPA mixture in an ultrasonic bath for 20 min.
5. Stir the CNT–IPA solution with a mechanical stirrer for 20 min.
6. Inspect the solution for large CNT agglomerates at the bottom. If present, manually break them up with a spatula/spoon and continue mechanical stirring for an additional 20 min.
7. Add silicone oil to the CNT–IPA mixture and sonicate for 20 min.
8. Stir the CNT–SO–IPA mixture mechanically for 20 min.
9. Add PDMS base to the mixture, then sonicate for 20 min.
10. Stir the CNT–SO–PDMS–IPA mixture mechanically for 20 min.
11. Inspect again for agglomerates. If any remain, manually disperse them and sonicate for an additional 20 min.
12. Heat the precursor mixture at 55 °C under continuous stirring with the parafilm removed. Adjust stirring speed as needed to prevent splashing. Maintain stirring for ~2 h until the mixture thickens. A protective barrier around the beaker is recommended.
13. Continue heating for ~24 h. If no liquid phase remains, stop mechanical stirring and heat for an additional 24 h with stirring twice with spoons to ensure complete IPA evaporation. The precursor is ready when no IPA odor is detectable.
14. Take around 1g of CNT-PDMS-SO precursor within a small plastic mixing cup.

15. Add PDMS curing agent into the precursor with 10:2 weight ratio, 10 for CNT-PDMS-SO precursor and 2 for curing agent, weighting on the scale, and mixing in small mixing cup.
16. Cut off 1cm length of copper strip, fold it in on itself to form 0.5x2cm copper filament.
17. Cut the mount tape or red tape (~1.2 mm thickness) and stick them onto the glass slides to form a square reservoir for the CNT-PDMS-SO precursor.
18. Place the copper filament on top of one wall of the reservoir, then add a second layer of tape overtop. The filament will then be pinned down in between two layers of tape with part of it sticking into the center of the reservoir.
19. Put the CNT-PDMS-SO precursor on the glass slide and use a razor blade to smear and make it flat in the reservoir.
20. Put the CNT-PDMS-SO precursor inside of the vacuum chamber for 30 minutes for de-bubbling.
21. Add more CNT-PDMS-SO precursor on the glass slide and use the razor blade to smear and make it flat again.
22. Put the CNT-PDMS-SO precursor back inside of the vacuum chamber for another 30 minutes of de-bubbling.
23. Use the razor blade to make it flat again.
24. Remove the three walls of tape that do not have the copper filament in between them from the glass slide, leaving only the CNT-PDMS-SO remaining with the copper filament inside of it.
25. Stick the precursor glass slide on the surface of the heater with double sided tape for VCDP procedure.

Fabrication of sugar molds

1. Keep spandex gloves on, put heat resistant glove on non-dominant hand and have small spoon ready in dominant hand.
2. Heat the sugar (Redpath) at 208°C-210°C in a petri dish with a second (larger) petri dish over top as a cover and wait for it to melt until it turns fully liquid and just starts to bubble.
3. Start heating the soft mold in another petri dish alongside so that it is hot when sugar is ready.
4. Take the small spoon and mix melted sugar well so all sugar melts evenly, once it is light brown and just starting to bubble, take off the petri dish used as a lid, then take a bit of the liquid using the small spoon and pour a few drops onto the mold carefully. Note: the mold should already be heated on its own hotplate at this point.
5. Use the scalpel to remove the air bubbles inside the soft mold and be careful not to break the soft mold. Note: this part may take some time, and you may need to get

more drops from the melted sugar periodically to reheat the sugar on the mold, so it flows well into the holes. Also make sure to check beneath the mold to see if there are any hidden bubbles. Sometimes it also helps to pick up and fold the mold upward so that the holes are easier to fill without air bubbles.

6. Once no bubbles remain, place petri dish with soft mold inside into a large beaker filled with cold water to quickly cool of the sugar.
7. After the mold is cooled down, demold the sugar from the soft mold.
8. Keep the sugar mold in cool, dry environment without direct exposure to the air.

VCDP Procedure

1. Stick the sugar mold to the glass slide with Epoxy glue and adjust the placement.
2. Use tape to fix the glass slide on the x, y, z linear stage for the VCDP set up.
3. Adjust the microscope to clearly observe the interface between the CNT-PDMS-SO precursor and the sugar beads, ensuring they are completely parallel with one another.
4. Lower the sugar beads in z direction with the detailed linear stage so that they are slowly dipped into the precursor, with around 1/3 to 1/2 of their diameter going into the precursor.
5. Abruptly raise the sugar mold with the bottom lifting station in z direction to form the suction cup and pillar structures, accompany with microscope.
6. After the structure was formed successfully, heat up the precursor at 90C˚ for 24h.